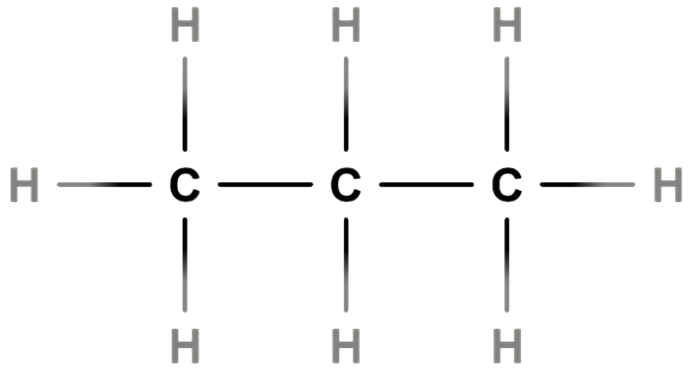
# 01 | PROPANE

**IUPAC Name:** Propane
**Molecular Formula:** $C_3H_8$

## 01 MOLECULAR FORMULA

$C_3H_8$

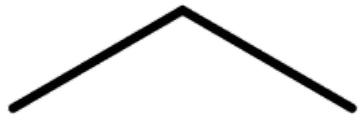
## 02 LEWIS STRUCTURE




## 03 CONDENSED STRUCTURAL FORMULA

$CH_3CH_2CH_3$

## 04 BOND LINE STRUCTURE



**Verification Strip**

- ☑ atom count verified
- ☑ valence checked
- ☑ bond order checked
- ☑ same connectivity confirmed

# 02 | ETHANOL

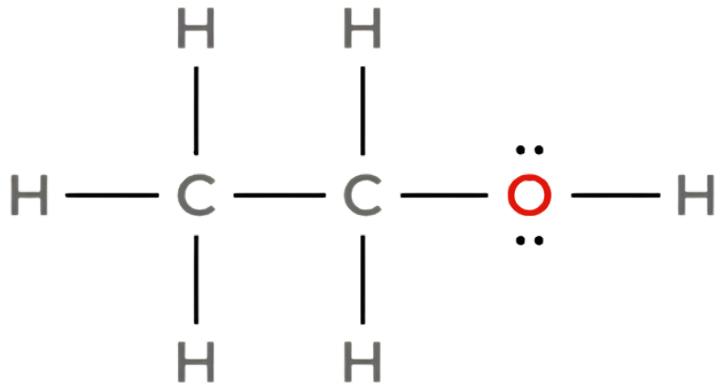
**IUPAC Name:** Ethanol

**Molecular Formula:** $C_2H_6O$

## 01 MOLECULAR FORMULA

$C_2H_6O$

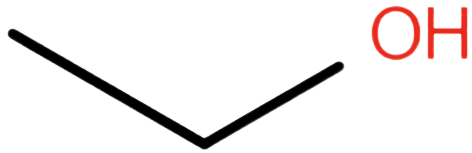
## 02 LEWIS STRUCTURE




## 03 CONDENSED STRUCTURAL FORMULA

$CH_3CH_2OH$

## 04 BOND LINE STRUCTURE




**Verification Strip**

- ☑ atom count verified
- ☑ valence checked
- ☑ bond order checked
- ☑ same connectivity confirmed

# 03 | Acetone

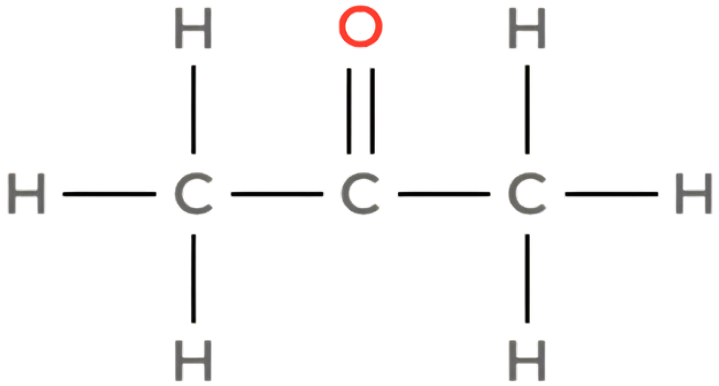
**IUPAC Name:** Propan-2-one

**Molecular Formula:** $C_3H_6O$

## 01 MOLECULAR FORMULA

$C_3H_6O$

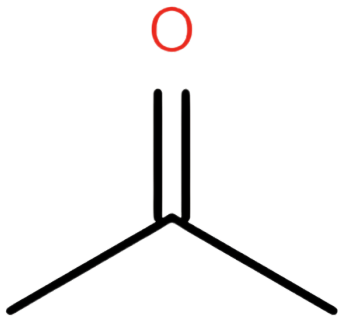
## 02 LEWIS STRUCTURE




## 03 CONDENSED STRUCTURAL FORMULA

$CH_3COCH_3$

## 04 BOND LINE STRUCTURE




**Verification Strip**

- ☑ atom count verified
- ☑ valence checked
- ☑ bond order checked
- ☑ same connectivity confirmed

# 04 | ACETIC ACID

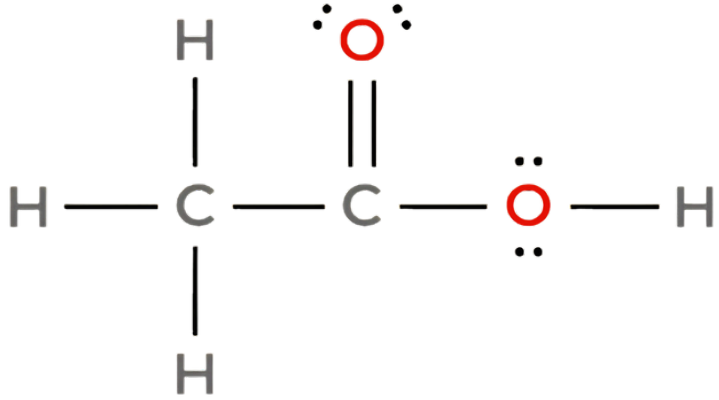
**IUPAC Name:** Ethanoic acid

**Molecular Formula:** $C_2H_4O_2$

## 01 MOLECULAR FORMULA

$$C_2H_4O_2$$

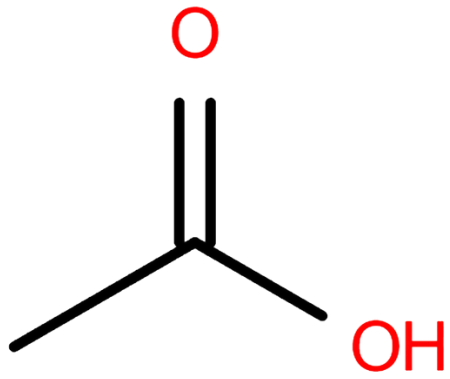
## 02 LEWIS STRUCTURE




## 03 CONDENSED STRUCTURAL FORMULA

$$CH_3COOH$$

## 04 BOND LINE STRUCTURE




**Verification Strip**

- ☑ atom count verified
- ☑ valence checked
- ☑ bond order checked
- ☑ same connectivity confirmed

# 05 | ETHYL ACETATE

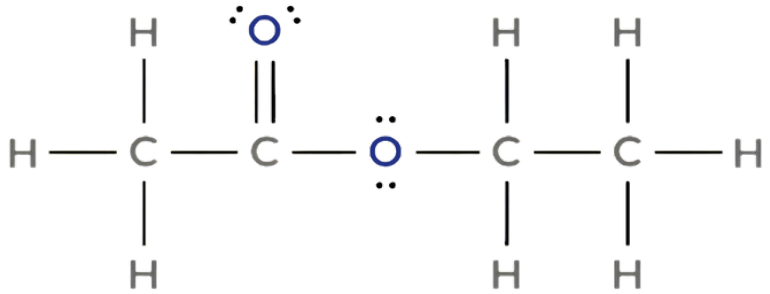
**IUPAC Name:** Ethyl ethanoate

**Molecular Formula**: $C_4H_8O_2$

## 01 MOLECULAR FORMULA

$$C_4H_8O_2$$

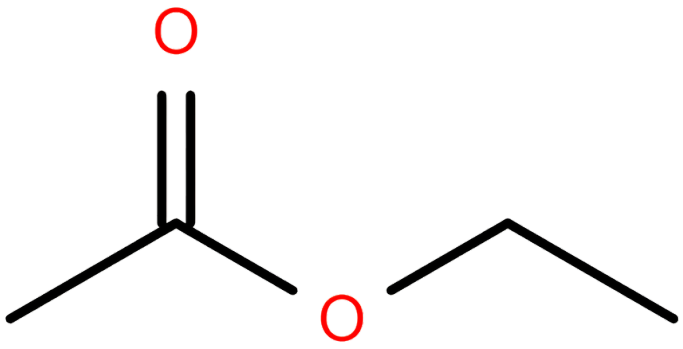
## 02 LEWIS STRUCTURE



## 03 CONDENSED STRUCTURAL FORMULA

$$CH_3COOCH_2CH_3$$

## 04 BOND LINE STRUCTURE




**Verification Strip**

- ☑ atom count verified
- ☑ valence checked
- ☑ bond order checked
- ☑ same connectivity confirmed

# 06 | CYCLOHEXANE

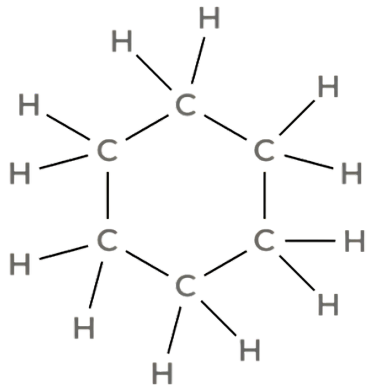
**IUPAC Name:** Cyclohexane

**Molecular Formula:** $C_6H_{12}$

## 01 MOLECULAR FORMULA

$$C_6H_{12}$$

## 02 LEWIS STRUCTURE

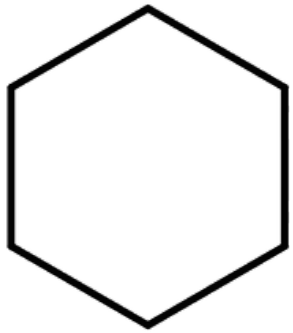



## 03 CONDENSED STRUCTURAL FORMULA

$$(CH_2)_6$$

*(cyclic; six $CH_2$ groups form a ring)*

## 04 BOND LINE STRUCTURE



**Verification Strip**

- ☑ atom count verified
- ☑ valence checked
- ☑ bond order checked
- ☑ same connectivity confirmed

# 07 | CYCLOHEXANONE

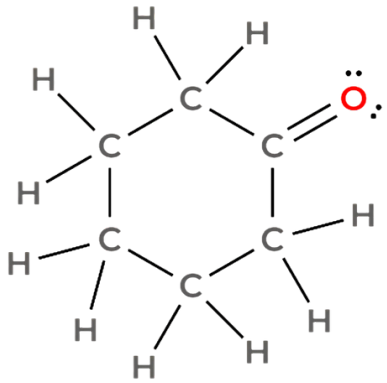
**IUPAC Name:** Cyclohexanone

**Molecular Formula:** $C_6H_{10}O$

## 01 MOLECULAR FORMULA

$$C_6H_{10}O$$

## 02 LEWIS STRUCTURE

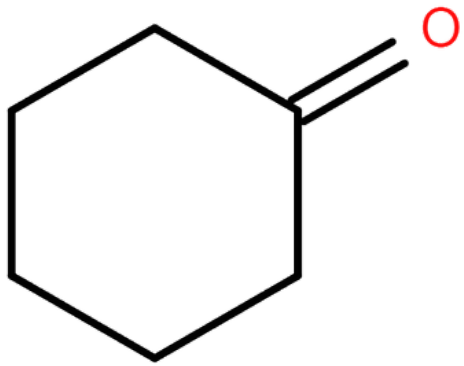



## 03 CONDENSED STRUCTURAL FORMULA

Cyclic $–CO–(CH_2)_5–$

*(ring containing one carbonyl carbon and five $CH_2$ groups)*

## 04 BOND LINE STRUCTURE




**Verification Strip**

- ☑ atom count verified
- ☑ valence checked
- ☑ bond order checked
- ☑ same connectivity confirmed

# 08 | ACETAMINOPHEN

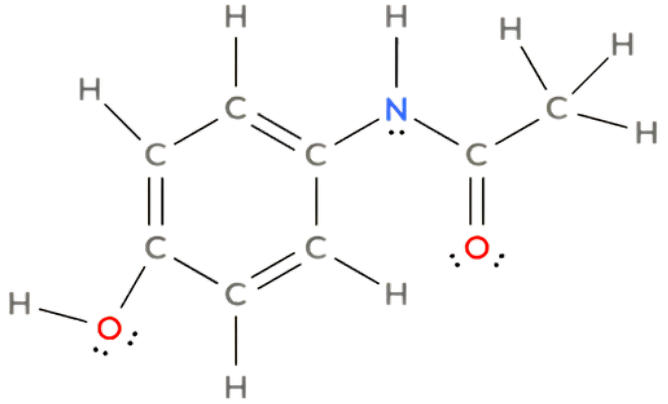
**Accepted Name:** Paracetamol | **IUPAC Name:** N-(4-hydroxyphenyl)acetamide

**Molecular Formula:** $C_8H_9NO_2$

## 01 MOLECULAR FORMULA

$$C_8H_9NO_2$$

## 02 LEWIS STRUCTURE

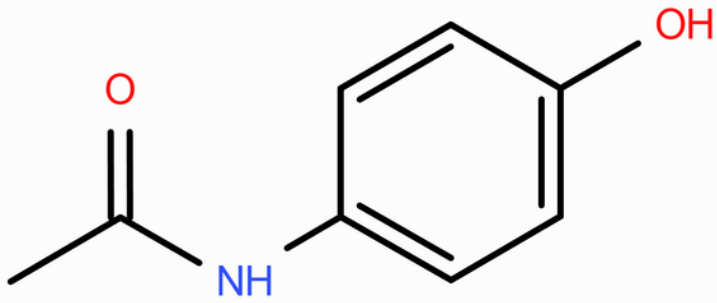



## 03 CONDENSED STRUCTURAL FORMULA

$$p\text{-}HOC_6H_4NHCOCH_3$$

*The "p-" indicates that the OH and $NHCOCH_3$ substituents are opposite each other on the benzene ring (para positions).*

## 04 BOND LINE STRUCTURE




**Verification Strip**

- ☑ atom count verified
- ☑ valence checked
- ☑ bond order checked
- ☑ same connectivity confirmed

# 09 | ASPIRIN

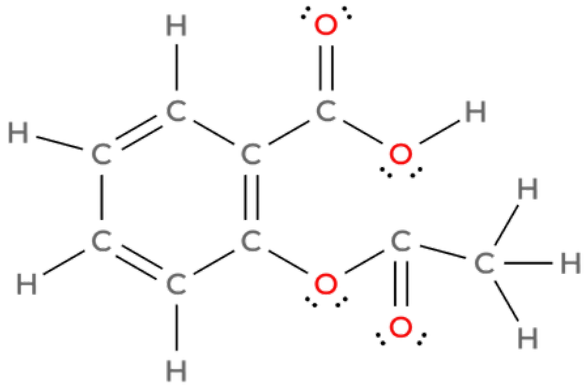
**Accepted Name:** Acetylsalicylic acid | **IUPAC Name:** 2-(acetyloxy)benzoic acid

**Molecular Formula:** $C_9H_8O_4$

## 01 MOLECULAR FORMULA

$$C_9H_8O_4$$

## 02 LEWIS STRUCTURE

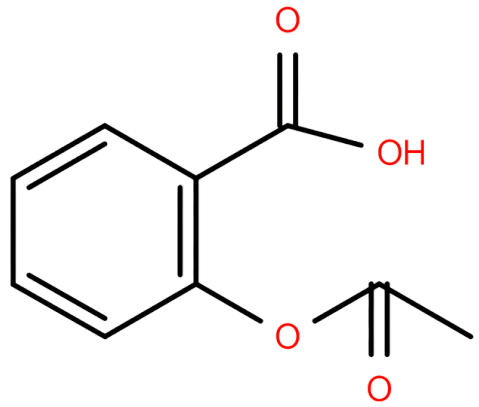


## 03 CONDENSED STRUCTURAL FORMULA

$$CH_3COOC_6H_4COOH$$

*(ortho-substituted benzene ring)*

## 04 BOND LINE STRUCTURE




**Verification Strip**

- ☑ atom count verified
- ☑ valence checked
- ☑ bond order checked
- ☑ same connectivity confirmed

# 10 | CAFFEINE

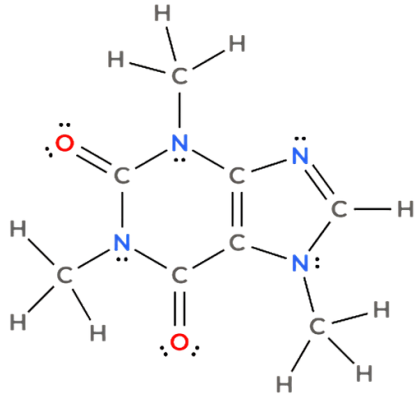
**IUPAC Name:** 1,3,7-trimethylpurine-2,6-dione

**Molecular Formula:** $C_8H_{10}N_4O_2$

## 01 MOLECULAR FORMULA

$$C_8H_{10}N_4O_2$$

## 02 LEWIS STRUCTURE

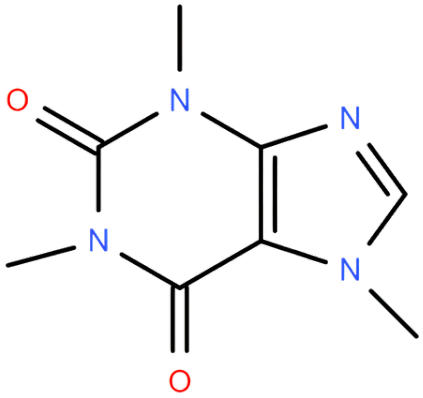


## 03 CONDENSED STRUCTURAL FORMULA

$$(CH_3)_3C_5HN_4O_2$$

*(Fused-ring structure; see bond-line representation.)*

## 04 BOND LINE STRUCTURE



**Verification Strip**

- ☑ atom count verified
- ☑ valence checked
- ☑ bond order checked
- ☑ same connectivity confirmed